

AZURE MONITORING

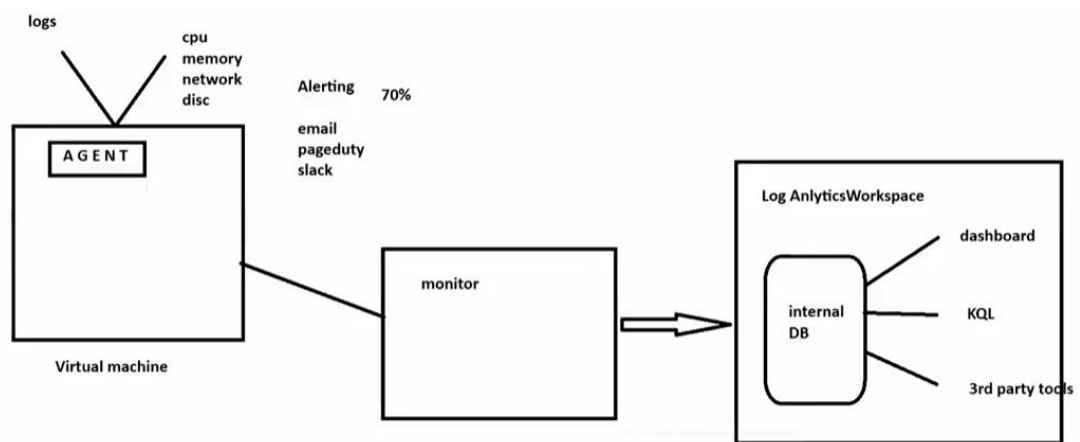
INTRODUCTION; Azure Monitor is a monitoring service in Microsoft Azure that collects, analyzes, and responds to telemetry data from Azure resources such as Virtual Machines.

It helps in performance monitoring, log analysis, and alert management.

Architecture Overview;

The monitoring architecture consists of the following components:

1. Virtual Machine (VM)
2. Azure Monitor
3. Log Analytics Workspace



1. Virtual Machine and Agent

An Azure Virtual Machine is installed with a monitoring agent.

The agent collects the following data:

- CPU usage
- Memory usage
- Network activity
- Disk performance
- System and application logs

This data is sent to Azure Monitor for analysis.

2. Azure Monitor

Azure Monitor receives telemetry data from the virtual machine.

It performs the following functions:

- Analyzes performance metrics

- Evaluates alert conditions (Example: CPU usage > 70%)
- Sends alerts via Email, Slack, or other notification services

3. Log Analytics Workspace

Log Analytics Workspace stores all collected log data.

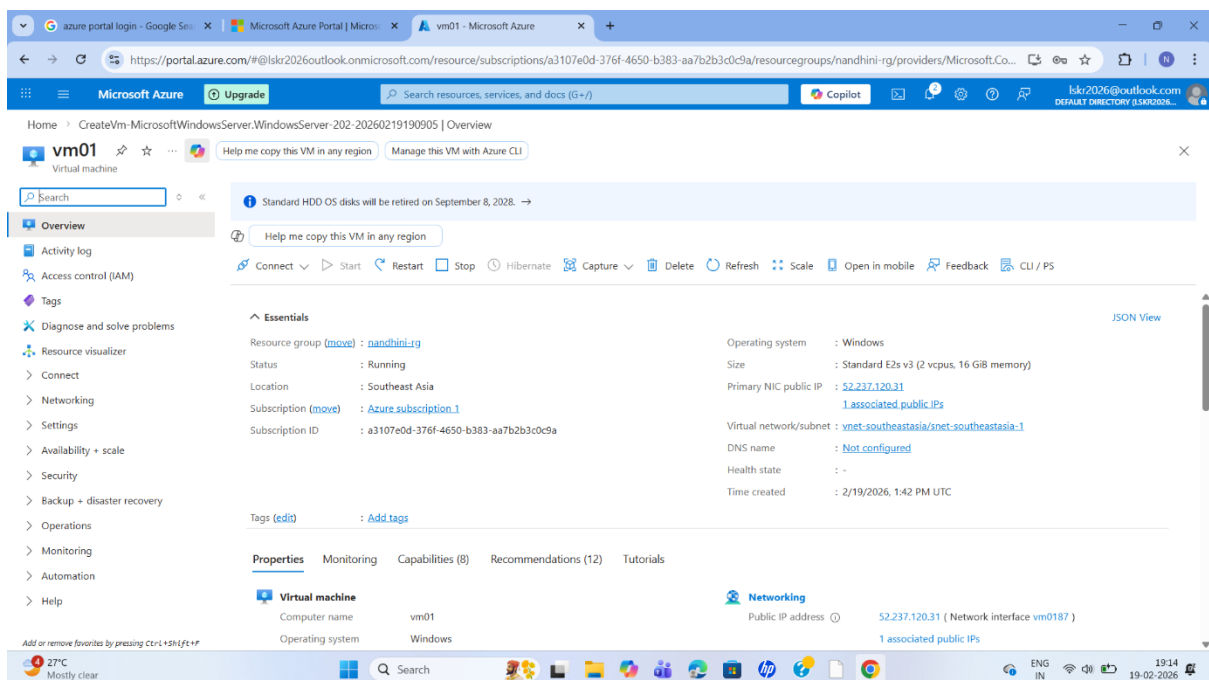
Features:

- Internal database for log storage
- KQL (Kusto Query Language) for querying logs
- Dashboards for visualization
- Integration with third-party tools

Users can write KQL queries to analyze performance and troubleshoot issues.

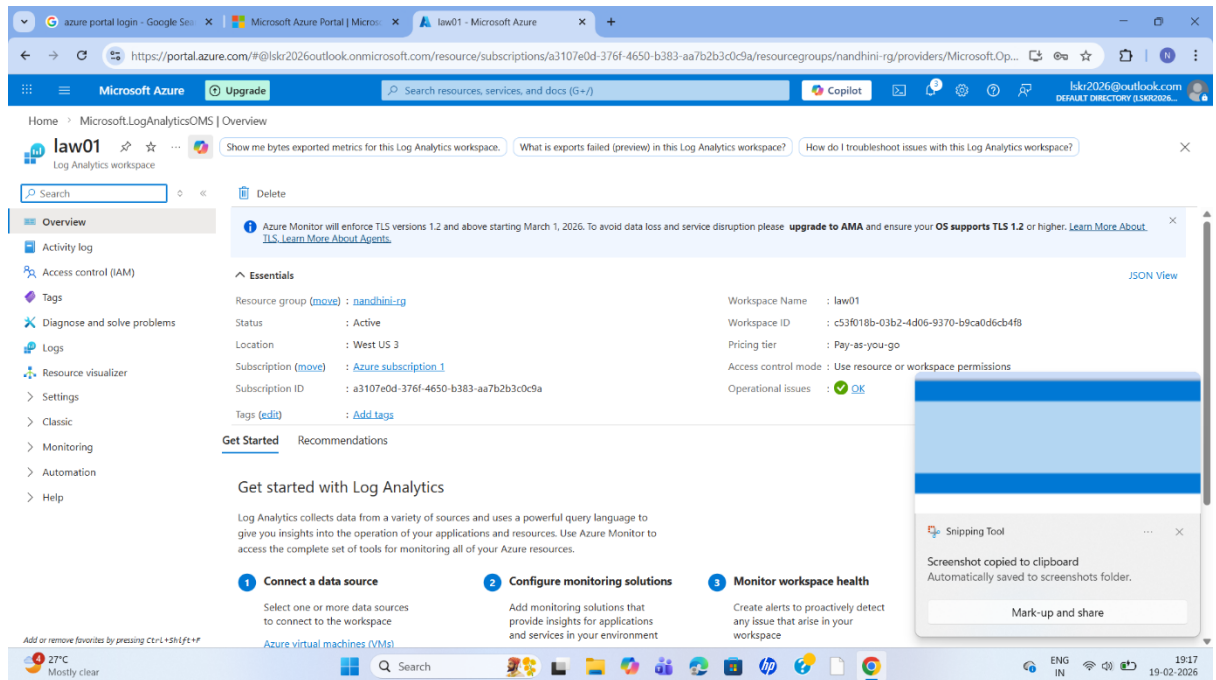
Step 1: Create Virtual Machine;

1. Go to Virtual Machines.
2. Click Create → Azure Virtual Machine.
3. Enter VM name, region, image, size.
4. Configure username and password.
5. Click Review + Create → Create



Step 2: Create Log Analytics Workspace

1. Login to Azure Portal.
2. Search for Log Analytics Workspace.
3. Click Create.
4. Select Subscription and Resource Group.
5. Provide Workspace Name.
6. Click Review + Create → Create.



Step 3: Enable Monitoring

1. Open the created VM.
2. Go to Monitoring section.
3. Click Insights → Enable.
4. Select the created Log Analytics Workspace.
5. Click Enable.

Azure Monitor Agent will be installed automatically.

azure portal login - Google S... Microsoft Azure Portal | Micro... Monitor - Microsoft Azure

https://portal.azure.com/#view/Microsoft_Azure_Monitoring/AzureMonitoringBrowseBlade/~:/virtualMachines

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Home Monitor

Monitor | Virtual Machines

Search Refresh Provide Feedback

Overview Activity log Alerts Issues (preview) Metrics Logs Change Analysis Service health Workbooks Dashboards with Grafana Insights Applications Virtual Machines Storage accounts Containers Networks Azure Cosmos DB

Get started Overview Performance Map

Monitor the health and performance of virtual machines

VM insights monitors the performance and health of your virtual machines and virtual machine scale sets, including their running processes and dependencies on other resources. It can help deliver predictable performance and availability of vital applications by identifying performance bottlenecks and network issues. [Learn more](#)

Enable VM Insights

Implement complete monitoring of your Azure and hybrid virtual machine environment. [Learn more](#)

[Configure Insights](#)

Analyze data

Analyze the health and performance for a single machine or multiple machines and drill into logs for troubleshooting. [Learn more](#)

[Analyze data](#)

Create alerts

Alerts in Azure Monitor proactively notify you of interesting data and patterns in your monitoring data and potentially take automated actions based on triggers. [Learn more](#)

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azure portal login - Google S... Microsoft Azure Portal | Micro... Monitor - Microsoft Azure

https://portal.azure.com/#view/Microsoft_Azure_Monitoring/AzureMonitoringBrowseBlade/~:/virtualMachines

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Get started Overview Performance Map

Filter by name... Subscription: Azure subscription 1 Resource group: All resource groups Type: All types Location: All locations

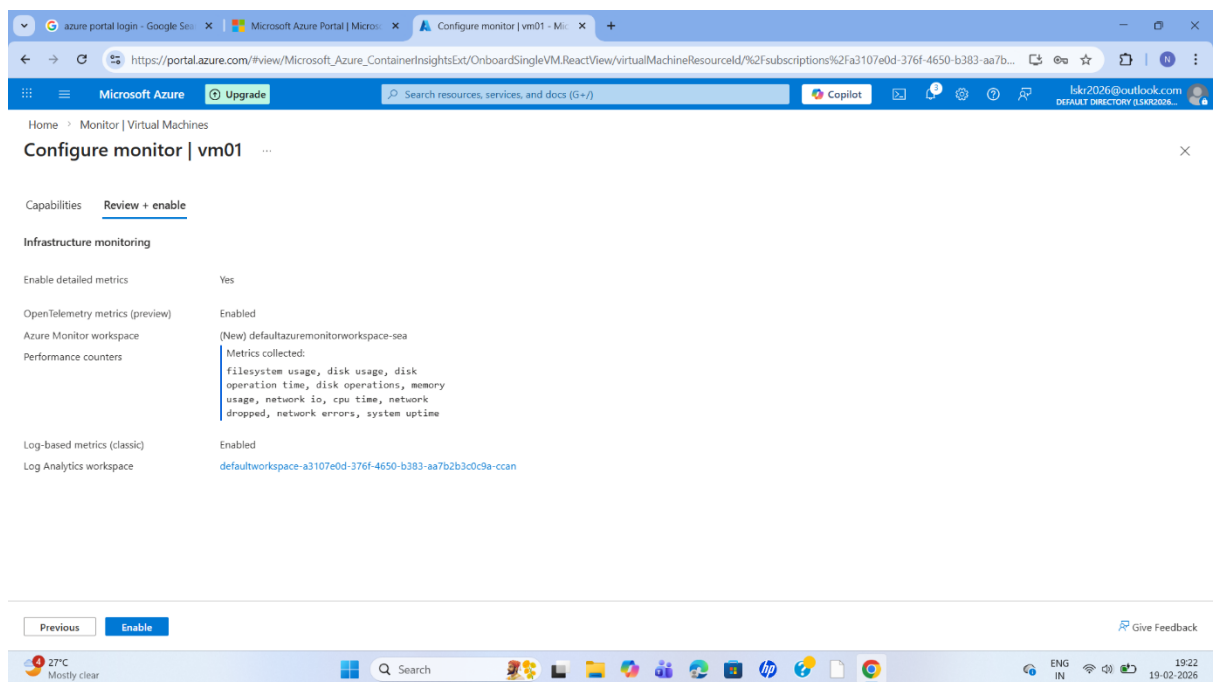
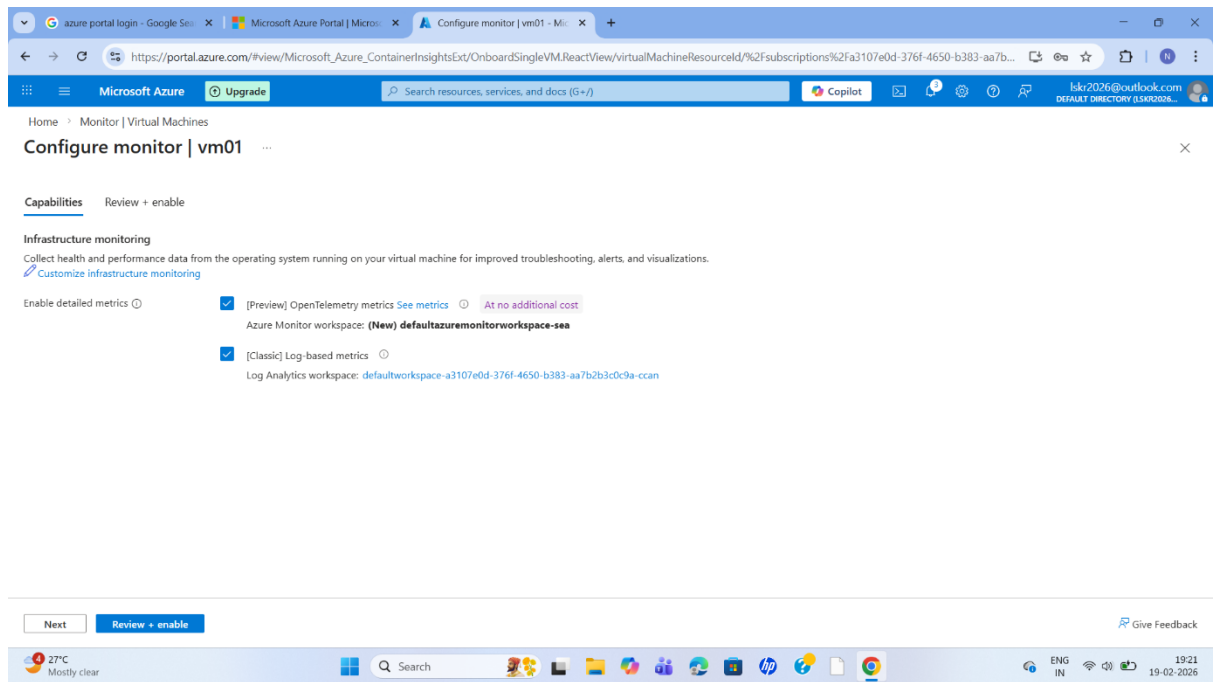
Group by: Subscription, Resource group

Onboarding now supports OpenTelemetry metrics. If you enable only OpenTelemetry metrics, your VM will still collect data but will not appear as 'Monitored' in this view. [Learn more](#)

Monitored (0) Not monitored (1) Workspace configuration Other onboarding options

Name	Monitor Coverage	Workspace
Azure subscription 1	1 of 1	
nandhini-rg	1 of 1	
vm01	Not enabled Enable	

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Step 4: Create Storage Account

1. Search → **Storage Accounts**
2. Click **Create**
3. Select:
 - Subscription
 - Resource Group
 - Storage Account Name (example: monitorstorage01)
 - Region (same as VM)
4. Performance → Standard

5. Click **Review + Create** → **Create**

✓ Storage Account created successfully.

The screenshot shows the 'Create a storage account' page in the Microsoft Azure portal. The 'Basic' tab is selected. The page includes the following fields and options:

- Subscription:** Azure subscription 1
- Resource group:** nandhini-rg (with a 'Create new' link)
- Instance details:**
 - Storage account name:** nandhini1234
 - Region:** (US) West US 3 (with a 'Deploy to an Azure Extended Zone' link)
 - Preferred storage type:** Choose preferred storage type (with a help message: 'This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. Learn more')
- Performance:**
 - ☒ **Standard:** Recommended for most scenarios (general-purpose v2 account)
 - ☐ **Premium:** Recommended for scenarios that require low latency.
- Redundancy:** Geo-redundant storage (GRS)

At the bottom, there are navigation buttons: 'Previous', 'Next', and 'Review + create'.

The screenshot shows the 'Create a storage account' page in the Microsoft Azure portal, with the 'Advanced' tab selected. The page includes the following sections and options:

- Basics:** (Previous tab)
- Advanced:** (Current tab)
- Networking:**
- Data protection:**
- Encryption:**
- Tags:**
- Review + create:**
- Security:**

Configure security settings that impact your storage account.

 - Require secure transfer for REST API operations:** ☒
 - Allow enabling anonymous access on individual containers:** ☒
 - Enable storage account key access:** ☒
 - Default to Microsoft Entra authorization in the Azure portal:** ☐
 - Minimum TLS version:** Version 1.2
 - Permitted scope for copy operations:** From any storage account
- Hierarchical Namespace:**

Hierarchical namespace, complemented by Data Lake Storage Gen2 endpoint, enables file and directory semantics, accelerates big data analytics workloads, and enables access control lists (ACLs) [Learn more](#)

 - Enable hierarchical namespace:** ☐

At the bottom, there are navigation buttons: 'Previous', 'Next', and 'Review + create'.

azure portal login - Google | azure portal login - Google | Microsoft Azure Portal | M | Create a storage account | Microsoft Azure Portal | Monitor - Microsoft Azure | +

https://portal.azure.com/#view/Microsoft_Azure_Storage/CreateStorageAccount.ReactView/_provisioningContext~/7B"initialValues"%3A%7B"subscriptionId"%3A%5B"a3107e0d-376f-46...

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Home > Storage center | Blob Storage

Create a storage account ...

Basics Advanced **Networking** Data protection Encryption Tags Review + create

Public access

Access your resource from anywhere through a public network.

Note: Allowing access to your resource through a public network increases security risk. [Learn more](#)

Public network access *

☒ Enable

Allow inbound and outbound access with the option to restrict select inbound access using resource access configurations for this resource.

☐ Disable

Restrict inbound access while allowing outbound access.

☐ Secure by perimeter (Most restricted)

Restrict inbound and outbound access using a network security perimeter. Secure by perimeter offers the greatest level of inbound and outbound restriction to secure your resource.

Public network access scope *

☒ Enable from all networks

☐ Enable from selected virtual networks and IP addresses

Enabling public network access will make this resource available publicly. Unless public access is required, consider using the most restricted access

Previous Next **Review + create**

Give feedback

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Search

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azure portal login - Google | azure portal login - Google | Microsoft Azure Portal | M | Create a storage account | Microsoft Azure Portal | Monitor - Microsoft Azure | +

https://portal.azure.com/#view/Microsoft_Azure_Storage/CreateStorageAccount.ReactView/_provisioningContext~/7B"initialValues"%3A%7B"subscriptionId"%3A%5B"a3107e0d-376f-46...

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Home > Storage center | Blob Storage

Create a storage account ...

Basics Advanced Networking **Data protection** Encryption Tags Review + create

Recovery

Protect your data from accidental or erroneous deletion or modification.

☐ Enable point-in-time restore for containers

Use point-in-time restore to restore one or more containers to an earlier state. If point-in-time restore is enabled, then versioning, change feed, and blob soft delete must also be enabled. [Learn more](#)

☐ Enable soft delete for blobs

Soft delete enables you to recover blobs that were previously marked for deletion, including blobs that were overwritten. [Learn more](#)

☐ Enable soft delete for containers

Soft delete enables you to recover containers that were previously marked for deletion. [Learn more](#)

Enabling soft delete for frequently overwritten data may result in increased storage costs [Learn more](#)

☐ Enable soft delete for file shares

Soft delete enables you to recover file shares that were previously marked for deletion. [Learn more](#)

Tracking

Manage versions and keep track of changes made to your blob data.

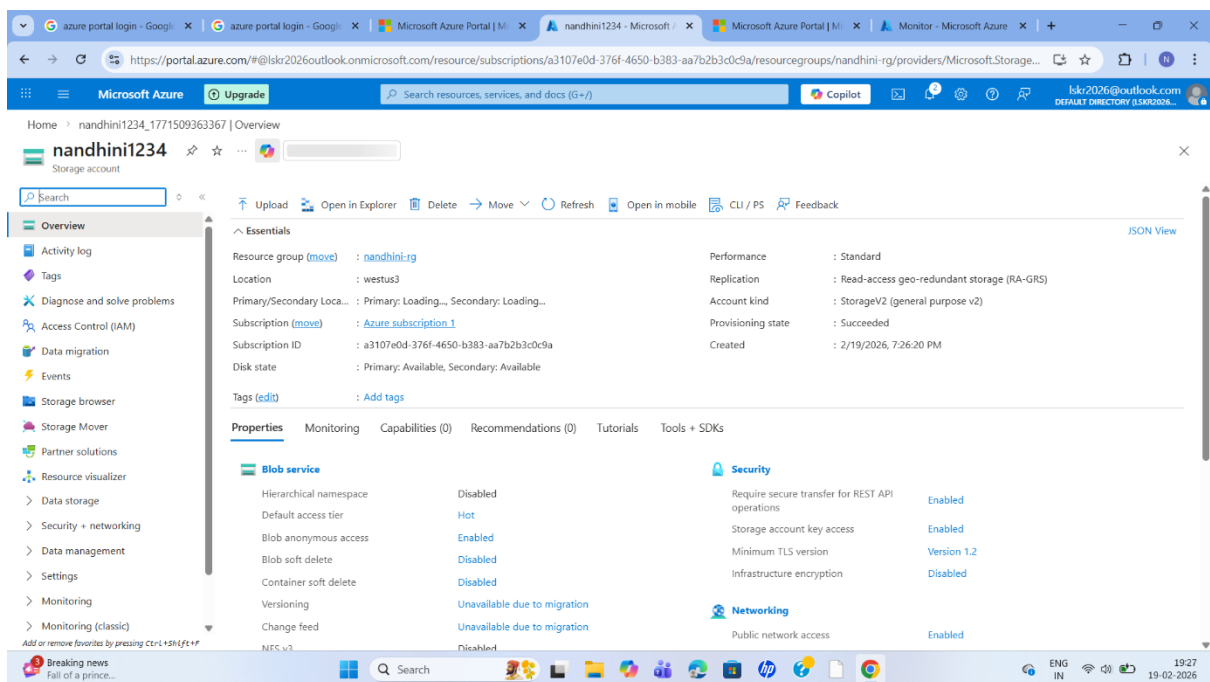
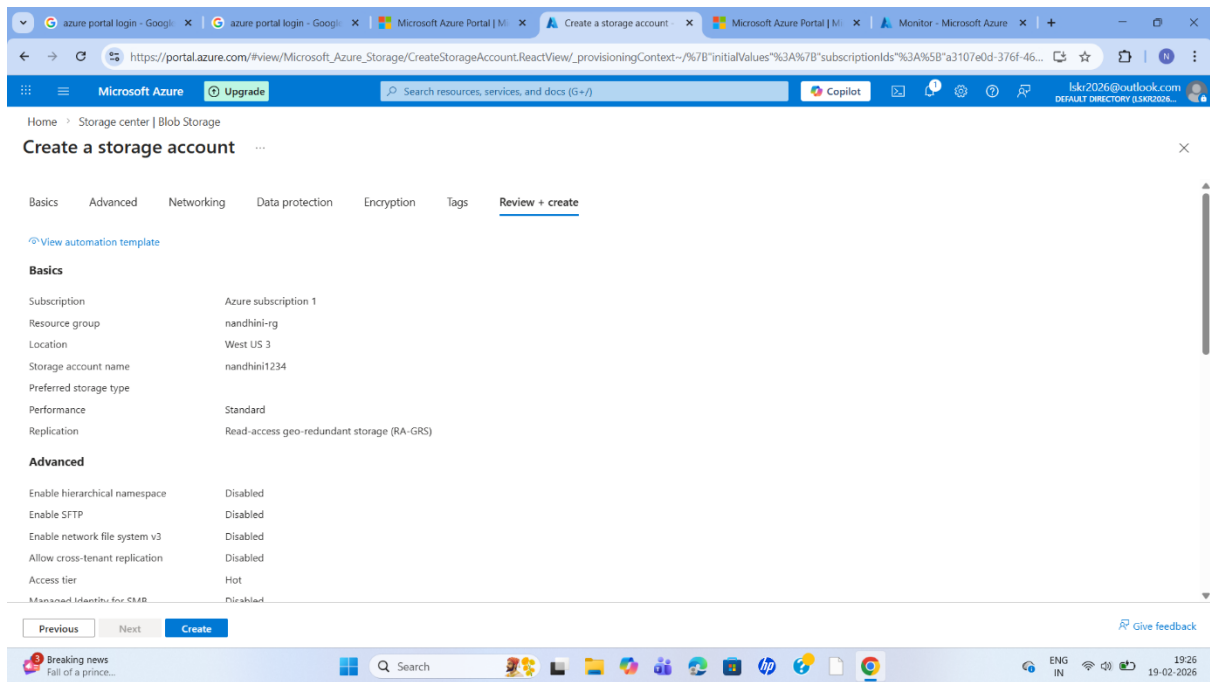
Previous Next **Review + create**

Give feedback

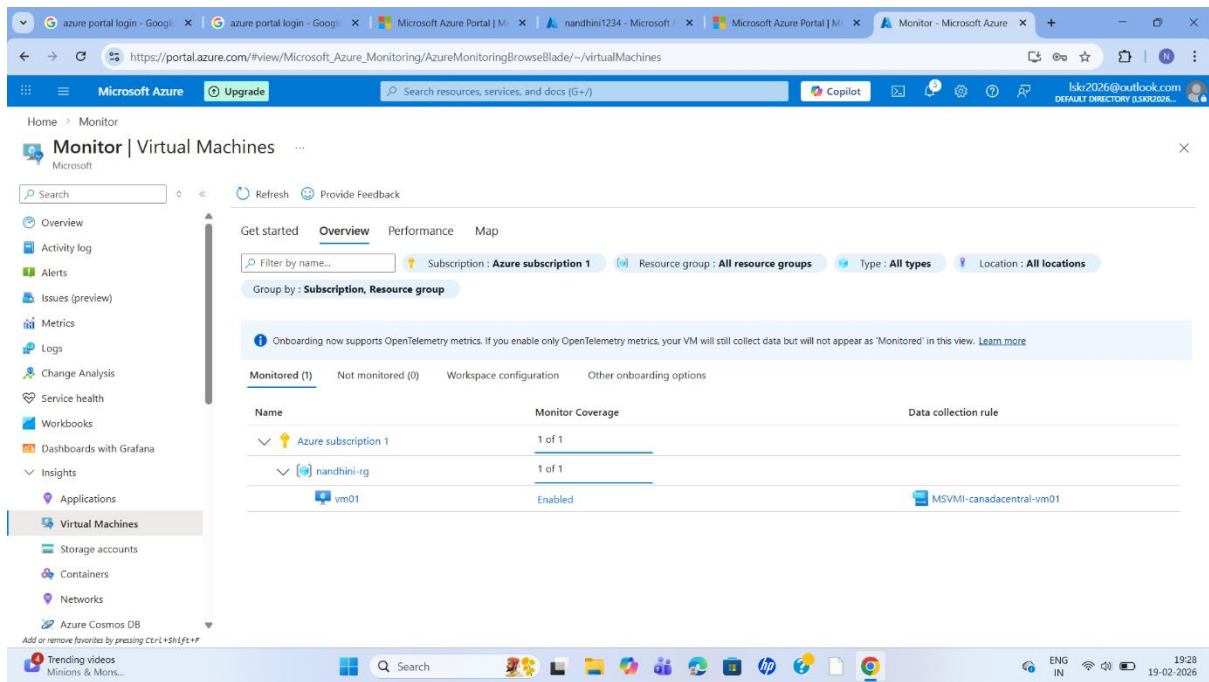
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Search

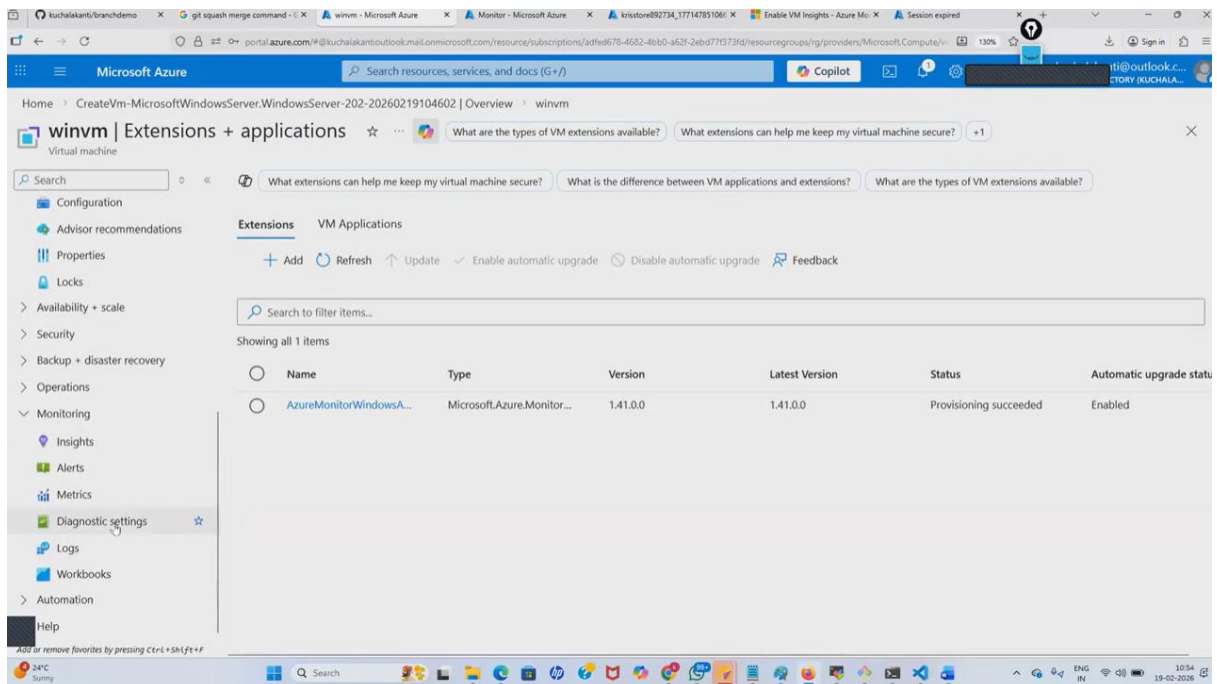
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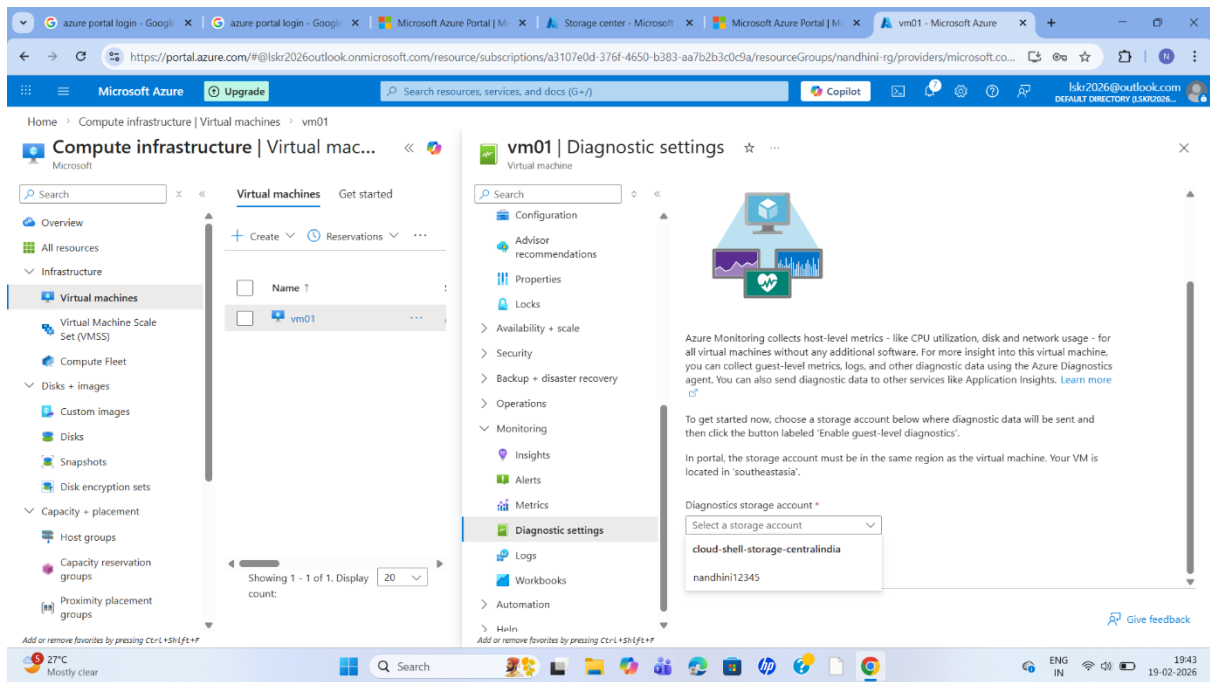
The virtual machine has been successfully added to azure monitor and monitoring is enabled, so this machine will be monitor by your azure monitor.



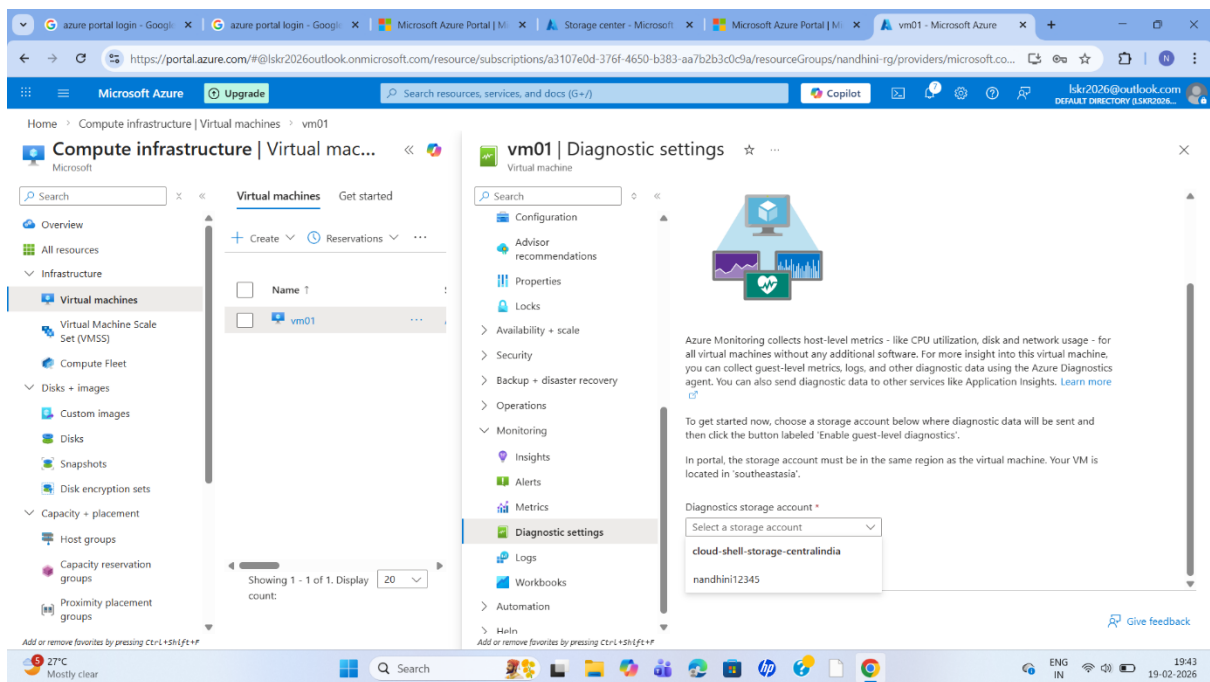
Again when i go back here, refresh



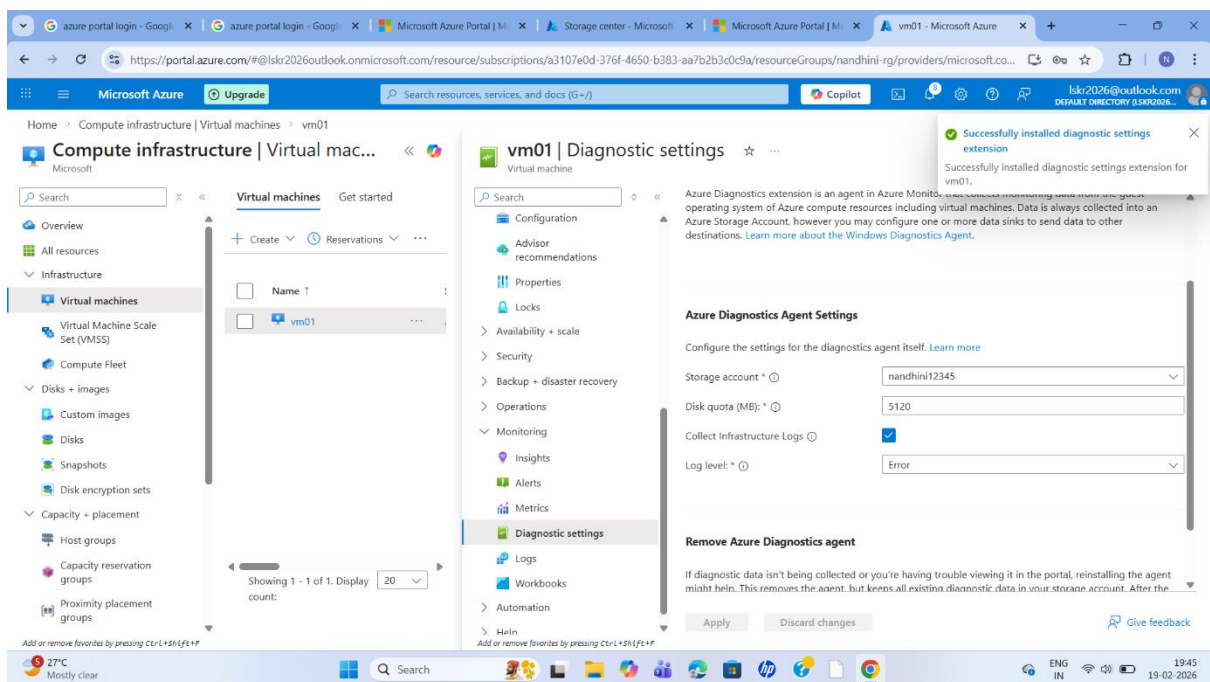
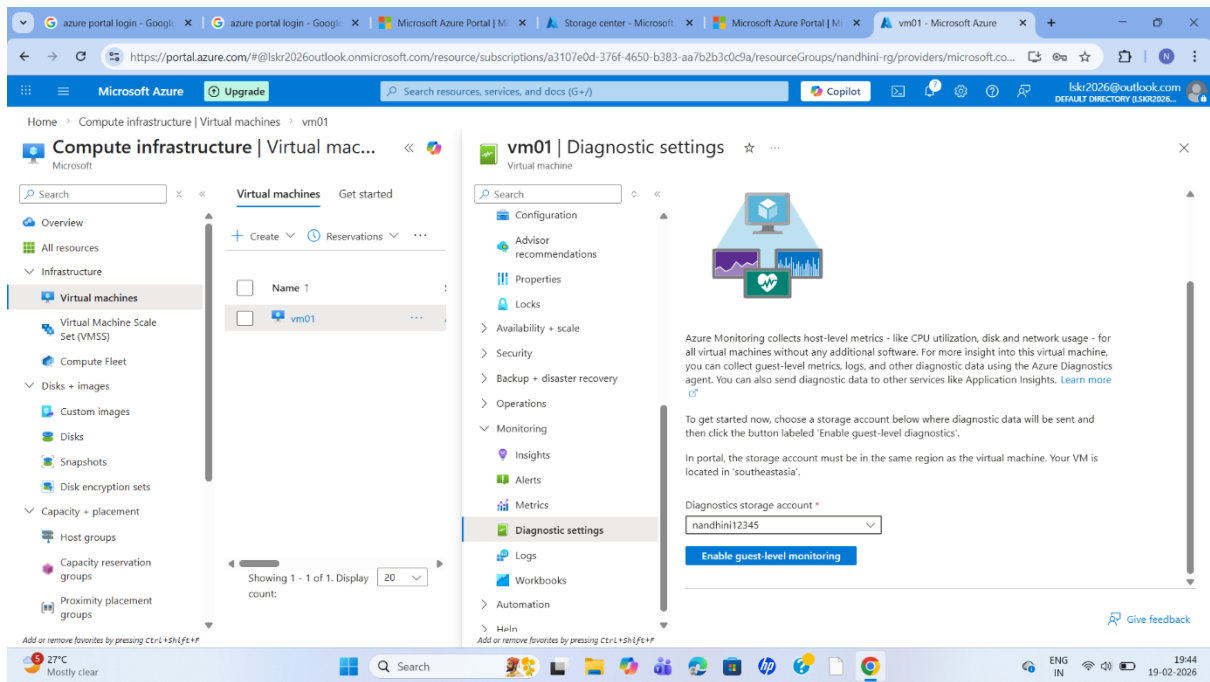
Diagnostics settings - virtual machine



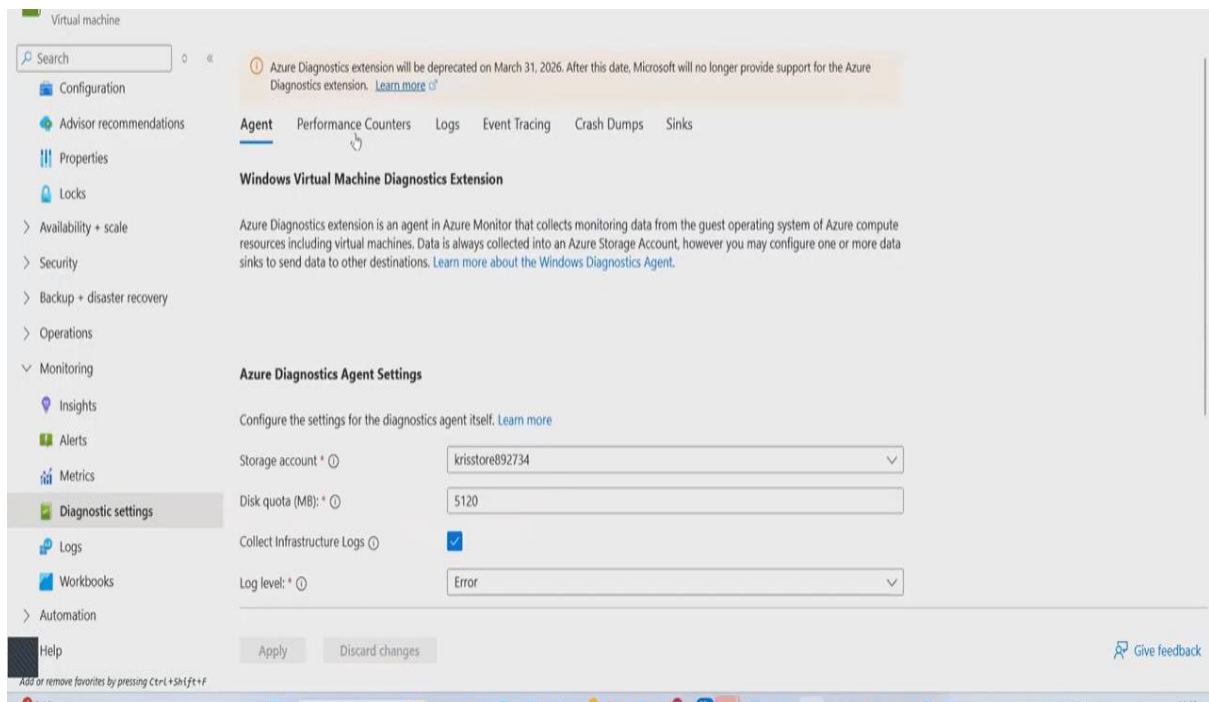
Select the storage account



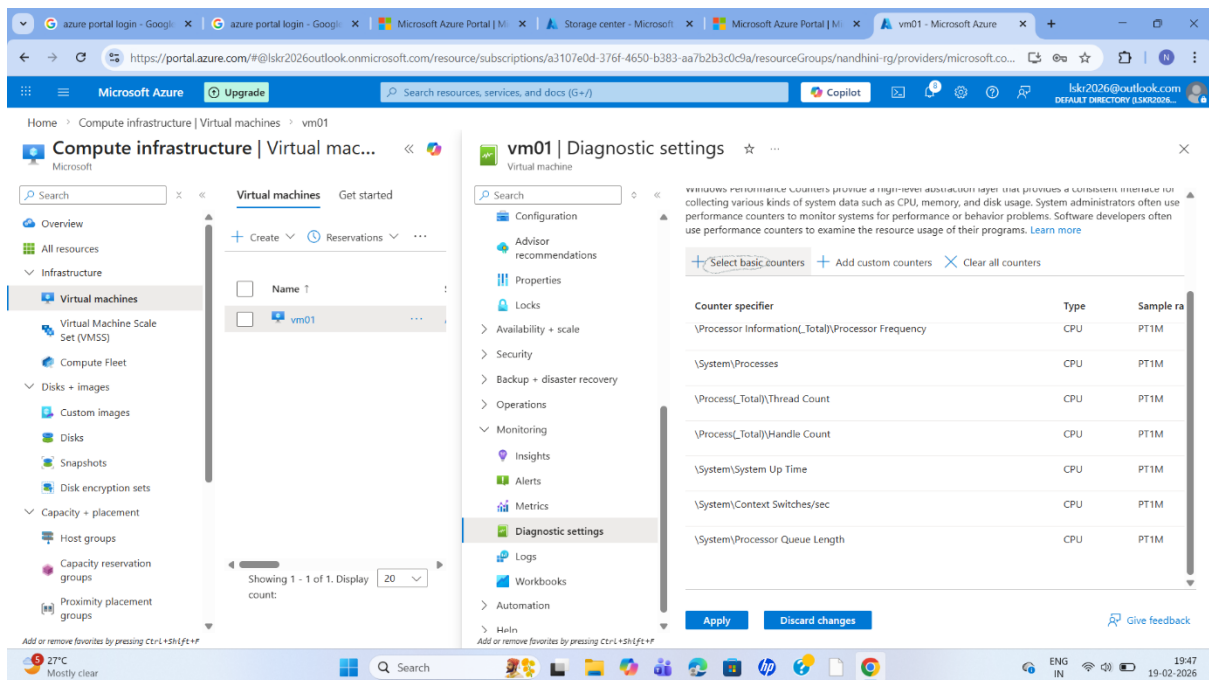
Enable guest level monitoring here.



And at the same time I am going to performance counter



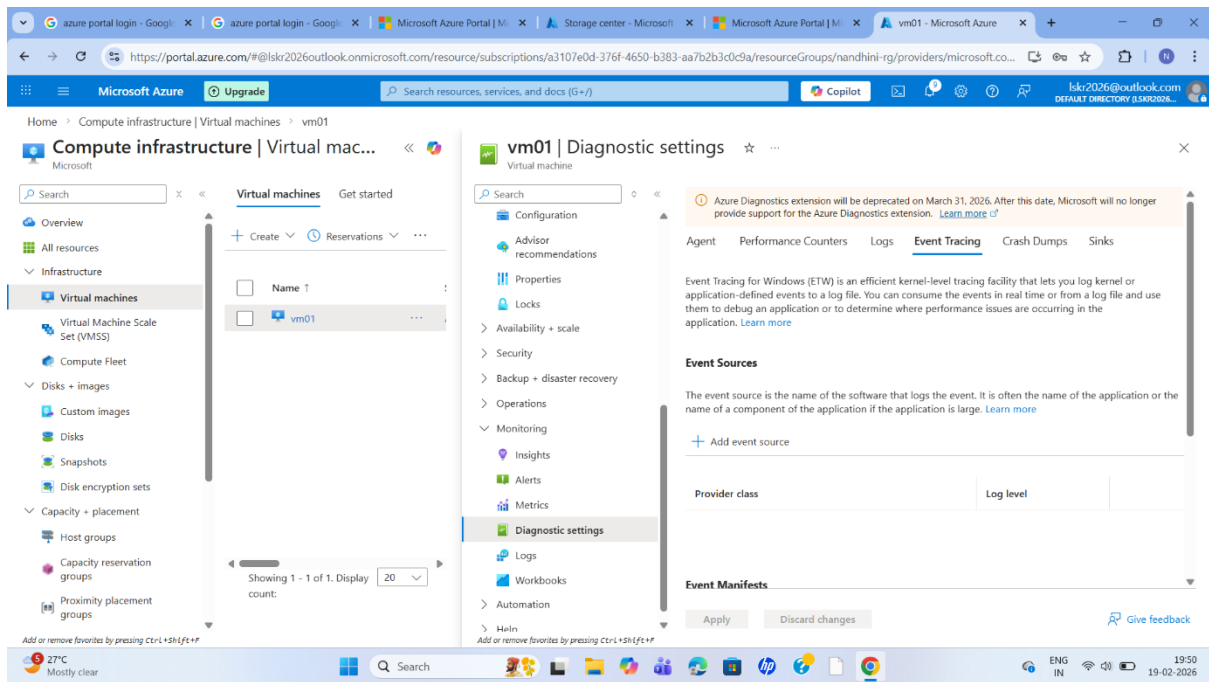
These are the counter it is going to add



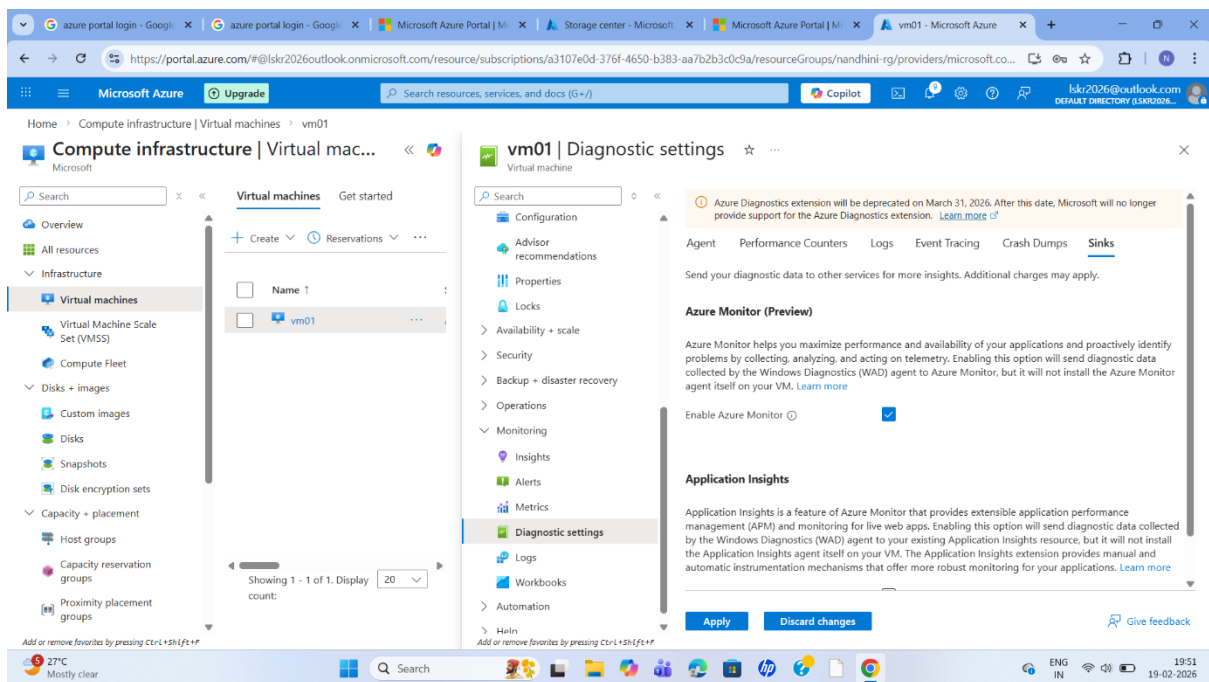
so if you going to select the basic counters

Microsoft Azure Portal interface showing the 'vm01 | Diagnostic settings' configuration page. The left sidebar displays the 'Virtual machines' section under 'Compute infrastructure'. The main content area shows the 'Diagnostic settings' for 'vm01', with the 'Metrics' tab selected. A dialog box titled 'Select basic counter groups' is open, allowing selection of performance counters. The selected counters are: CPU (10), Disk (18), Memory (10), Network (8), ASP.NET (140), and SQL Server (12). The sample rate is set to 1. The bottom status bar shows the time as 19:47 on 19-02-2026.

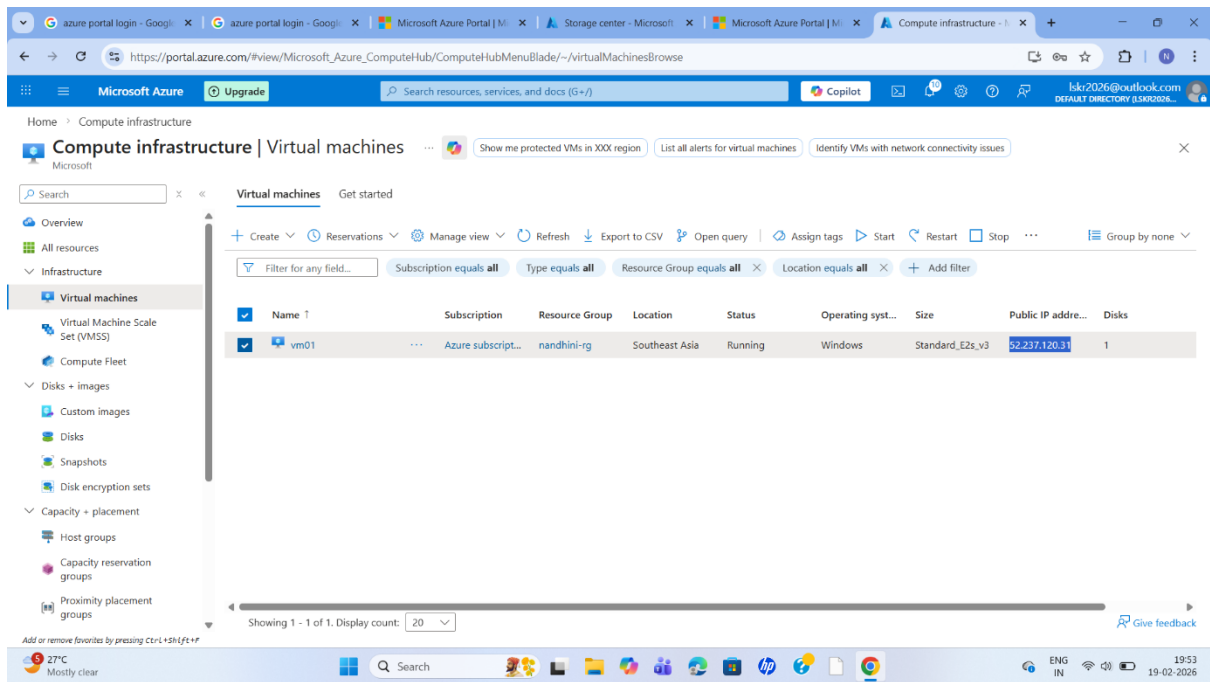
Microsoft Azure Portal interface showing the 'vm01 | Diagnostic settings' configuration page. The left sidebar displays the 'Virtual machines' section under 'Compute infrastructure'. The main content area shows the 'Diagnostic settings' for 'vm01', with the 'Logs' tab selected. A dialog box titled 'Select basic counter groups' is open, allowing selection of performance counters. The selected counters are: CPU (10), Disk (18), Memory (10), Network (8), ASP.NET (140), and SQL Server (12). The sample rate is set to 1. The bottom status bar shows the time as 19:50 on 19-02-2026.



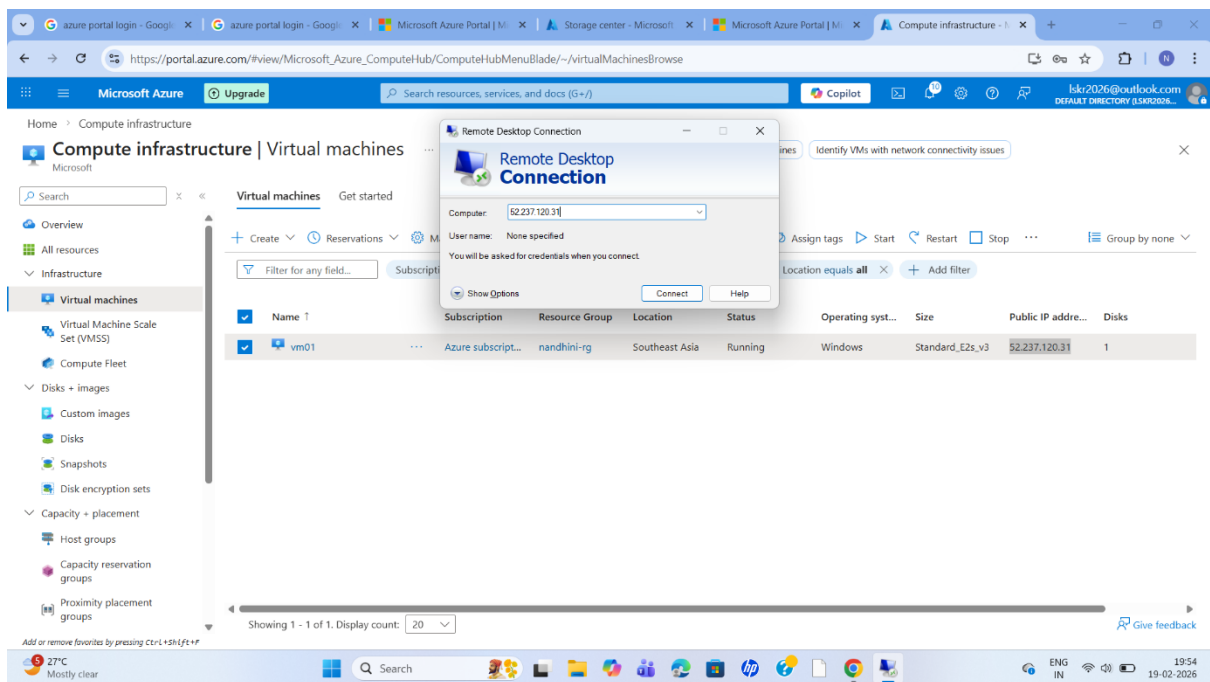
Now let me apply this,

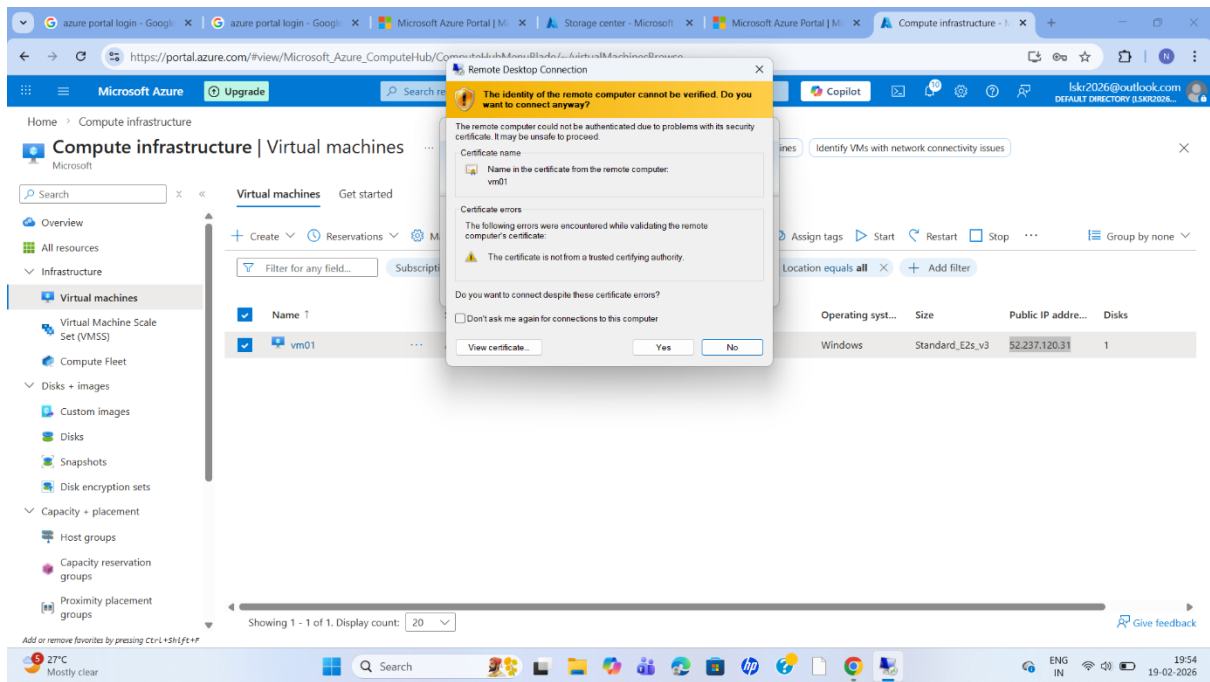


The public Ip address is assigned to the virtual machine.



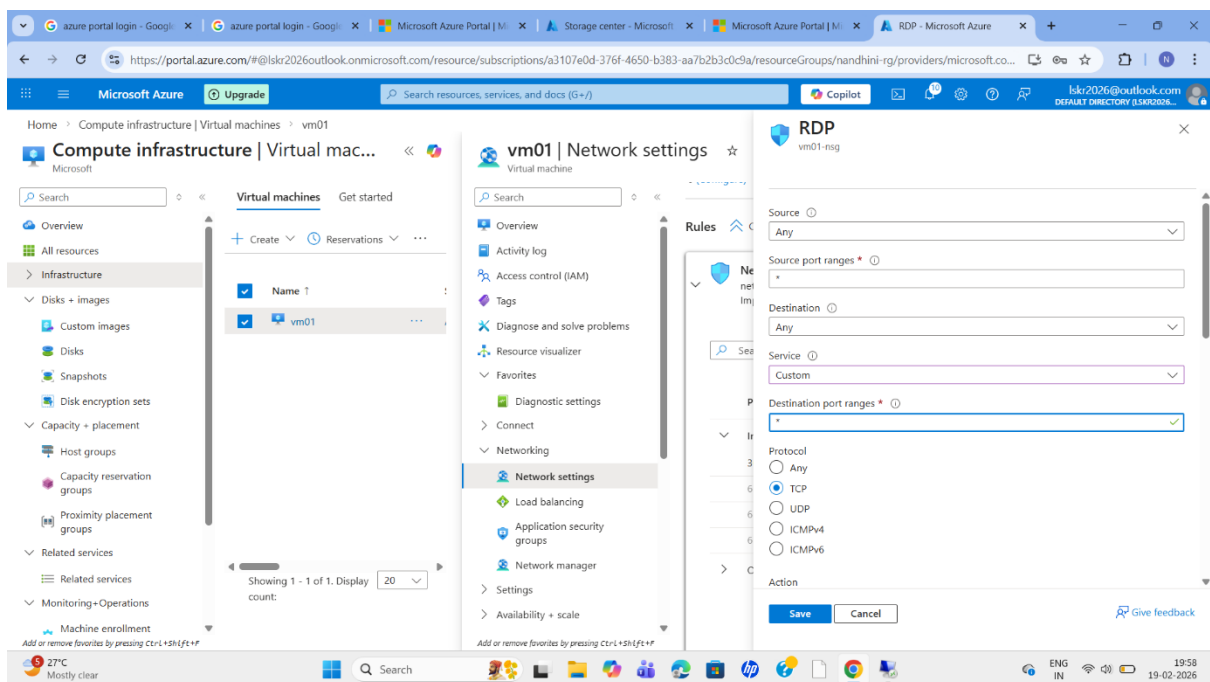
After that Remote Desktop Protocol (RDP) is used to connect to the windows virtual machine.

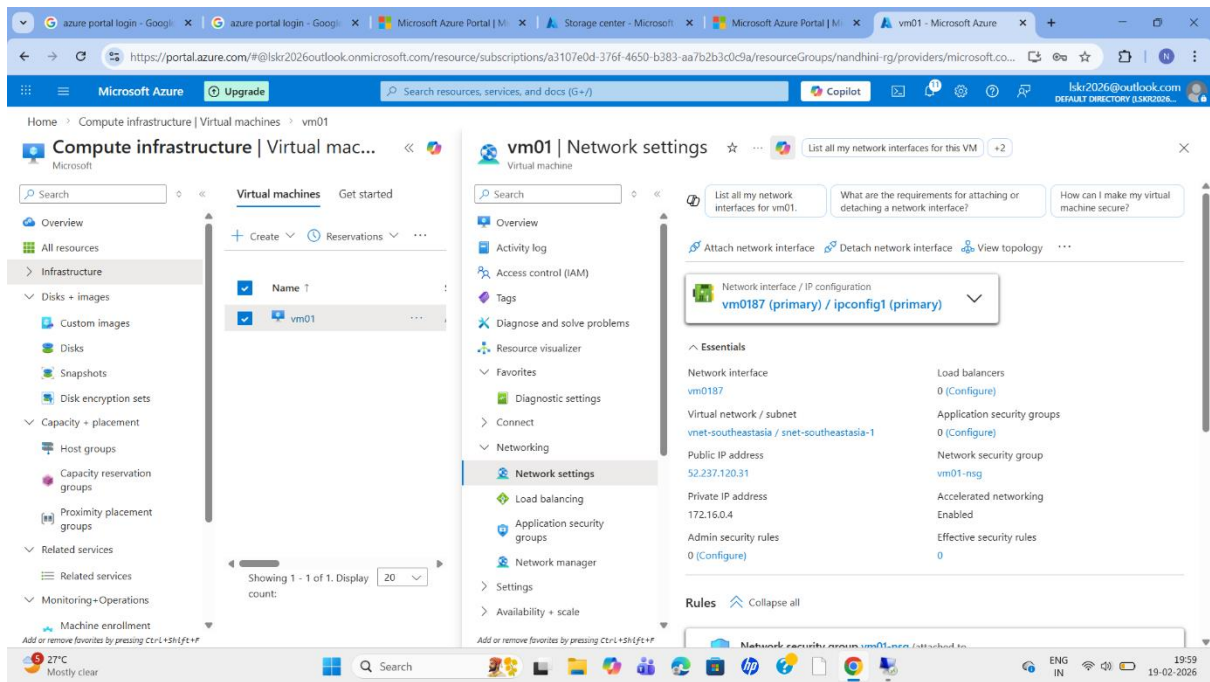




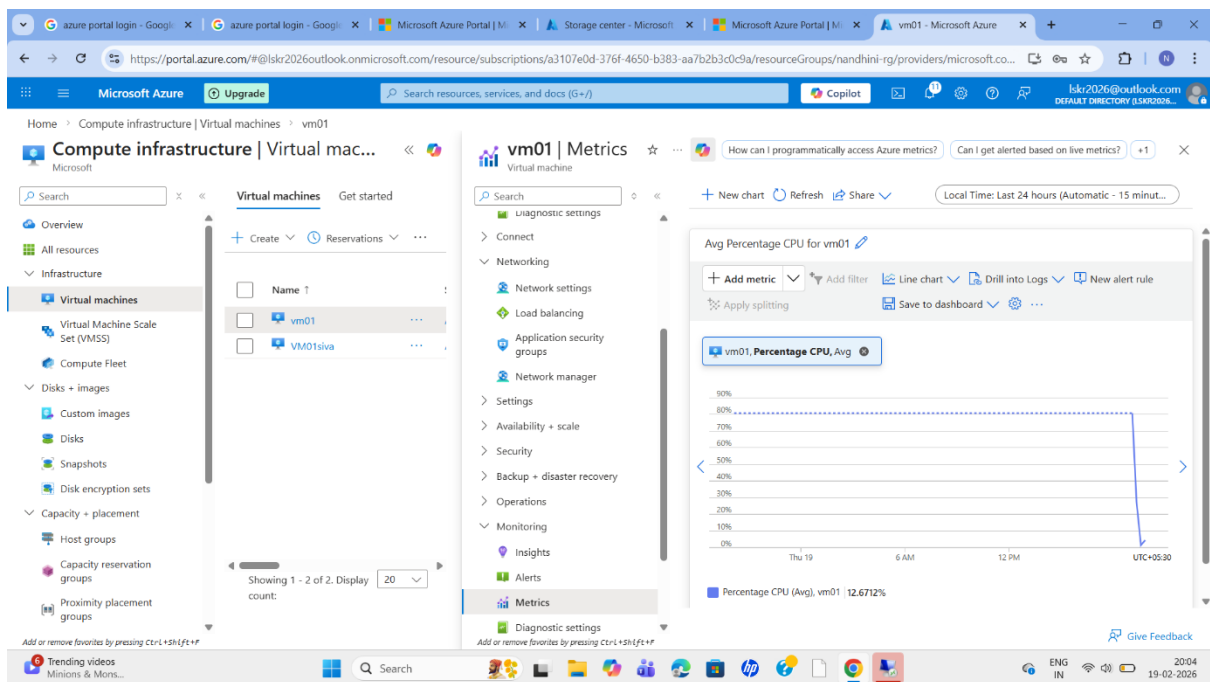
Configure an inbound security rule for remote desktop access

And save





Now I select a Metrics;



And save dashboard

And after create an alert rule;

Home > Compute infrastructure > Virtual machines > vm01 | Alerts

Create an alert rule

Scope **Condition** Actions Details Tags Review + create

Configure when the alert rule should trigger by selecting a signal and defining its logic.

Signal name * [See all signals](#)

Alert logic

Threshold type ☒ Static ☐ Dynamic

Aggregation type

Value is

Threshold * %

When to evaluate

Check every

[Review + create](#) [Previous](#) [Next: Actions >](#)

Preview \$0.10 USD/month

Whenever the average Percentage CPU is greater than 80%

Preview time range: Over the last 6 hours **Time series:** Aggregate

The chart displays a line representing CPU usage over a 6-hour period. The y-axis ranges from 10% to 80%. A red vertical line is drawn at the 80% mark, indicating the threshold. The line shows a sharp spike that crosses the threshold, followed by a drop and then a gradual decline.

Now create actions group

Home > Compute infrastructure > Virtual machines > vm01 | Alerts > Create an alert rule

Create action group

Basics Notifications Actions Tags Review + create

An action group invokes a defined set of notifications and actions when an alert is triggered. [Learn more](#)

Project details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription

Resource group * [Create new](#)

Region *

Instance details

Action group name *

Display name * The display name is limited to 12 characters

[Review + create](#) [Previous](#) [Next: Notifications >](#)

Next notification;

So I am going to only emails

So I am adding email here

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https://portal.azure.com/#view/Microsoft_Azure_Monitoring_Alerts/CreateActionGroupBlade/subscriptionId/a3107e0d-376f-4650-b383-aa7b2b3c0c9a/subscriptionName/resourceG...

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Home > Compute infrastructure > Virtual machines > vm01 | Alerts > Create an alert rule

Create action group

Basics Notifications Actions Tags Review + create

Choose how to get notified when the action group is triggered. This step is optional.

Notification type	Name	Selected
Email/SMS message...		

Review + create Previous Next: Actions >

Email/SMS message/Push/Voice

Add or edit Email/SMS message/Push/Voice action

☒ Email
Email *

☐ SMS (Carrier charges may apply)
Country code
Phone number

☐ Azure mobile app notification
Azure account email

☐ Voice
Country code
Phone number

Enable the common alert schema. [Learn more](#)
☐ Yes ☒ No

OK

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https://portal.azure.com/#view/Microsoft_Azure_Monitoring_Alerts/CreateActionGroupBlade/subscriptionId/a3107e0d-376f-4650-b383-aa7b2b3c0c9a/subscriptionName/resourceG...

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Home > Compute infrastructure > Virtual machines > vm01 | Alerts > Create an alert rule

Create action group

Basics Notifications Actions Tags Review + create

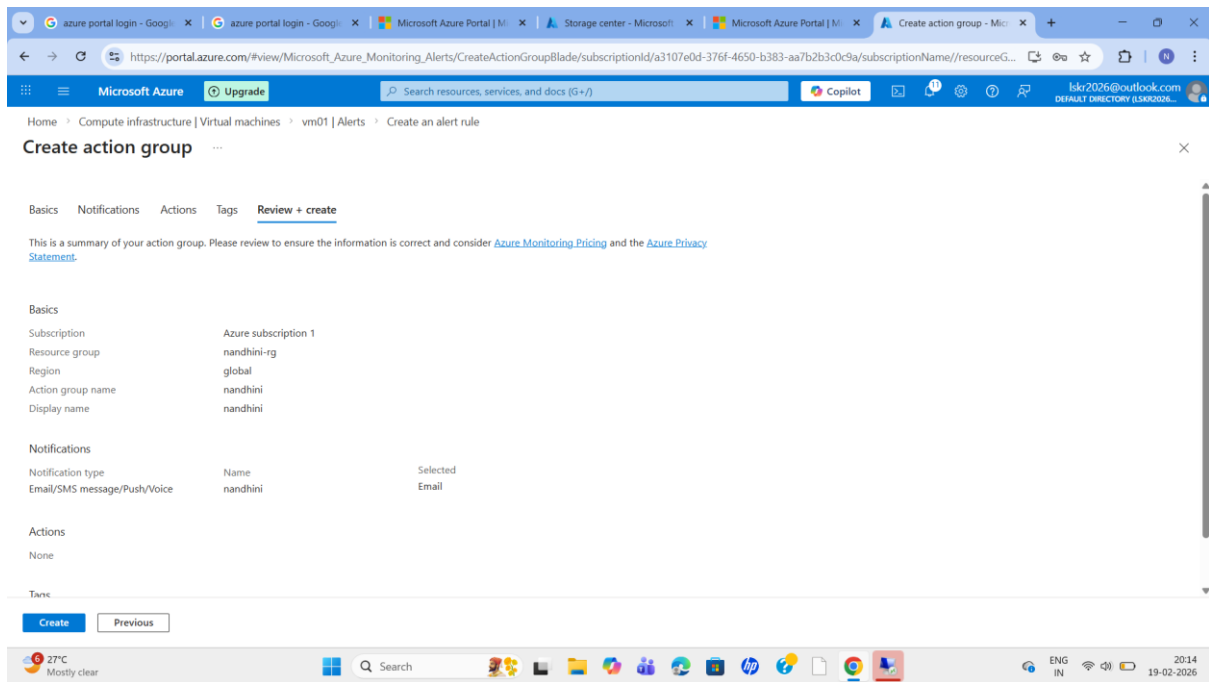
Choose which actions are performed when the action group is triggered. This step is optional.

Action type	Name	Selected	Identity

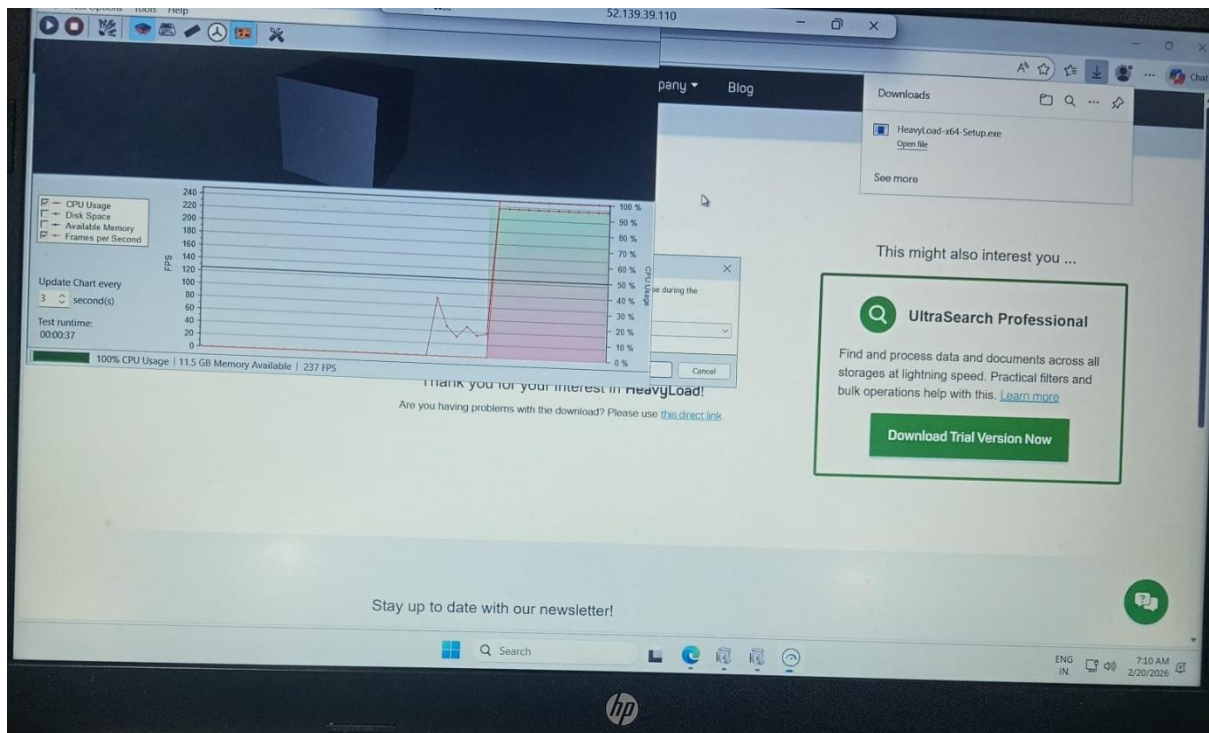
Review + create Previous Next: Tags >

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And create



Now go to our created vm which we logged in -open Microsoft edge-and download Stress software our vm-overview monitoring metrics



Here if it is <40, than email will be sent

Your Azure Monitor alert was triggered

Azure monitor alert rule nandhu was triggered for vm01 at February 20, 2026 7:17 UTC.

Alert rule description	nandhu
Rule ID	/subscriptions/05395d22-d4cd-42d0-931f-e8ceda209d2f/resourceGroups/nandhu-rg/providers/microsoft.insights/metricAlerts/nandhu View Rule >
Resource ID	/subscriptions/05395d22-d4cd-42d0-931f-e8ceda209d2f/resourceGroups/nandhu-rg/providers/Microsoft.Compute/virtualMachines/vm01 View Resource >

Your Azure Monitor alert was triggered

Azure monitor alert rule nandhu was triggered for vm01 at February 20, 2026 7:17 UTC.

Alert rule description	nandhu
Rule ID	/subscriptions/05395d22-d4cd-42d0-931f-e8ceda209d2f/resourceGroups/nandhu-rg/providers/microsoft.insights/metricAlerts/nandhu View Rule >
Resource ID	/subscriptions/05395d22-d4cd-42d0-931f-e8ceda209d2f/resourceGroups/nandhu-rg/providers/Microsoft.Compute/virtualMachines/vm01 View Resource >

